ADT25SOCB0...

MIT SCHOOL OF COMPUTING
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PUNE, INDIA
A Leap Towards World Class Education

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Subject: PL- Python Roll No.: 06 Exp. No.:
Name of the Experiment:
Performed on:
Submitted on:

Marks	Teacher's Signature with date
4 / 10	

Q.1] List the key features & advantages of python. Also mention any two python IDEs / editors.

Key features of python : 1. Simple & easy-to-learn syntax.
2. Interpreted programming language.
3. High-level language.
4. Object - oriented programming support.
5. Platform independent
6. Extensive standard library
7. Dynamic typing

Advantages of Python : 1. Reduces development time
2. Easy debbuing & maintenance
3. Support's multiple programming paradigms
4. Widely used in data science, AI, web development & automation

Python IDEs / editors :-
1. Pycharm
2. Visual studio code (VS code).

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Q.2] Explain the history & evolution of python. Describe why python is popular.

Python was created by Guido Van Rossum in 1989 & released in 1991. It was developed at the Centrum Wiskunde & Informatica (CWI). in the Netherlands. Python was designed to emphasize code readability & simplicity.

* Evolution of python :- ① Python 1.x :- Basic features
  ① Python 2.x : Improved performance & libraries.
  ① Python 3.x : Enhanced syntax, better Unicode support, & improved Security.

* Reasons for Python's Popularity : 1. Beginner-friendly syntax.
  2. Strong community support.
  3. Extensive libraries & frameworks.
  4. High demand in industry (AI, ML, data science, web development)
  5. Cross-platform compatibility.

Q.3] Program to accept two numbers & perform arithmetic operations

```
a = float (input (" Enter frist number : "))
b = float (input (" Enter secound number : "))

print (" Addition : ", a+b)
print (" Subtraction: ", a-b)
print (" Multiplication: " a*b)
print (" Division : ", a/b)
```

Q.4] Analyze the python code & identify data types & output.

* Data types of Variables :
  - a is of type int (integer)
  - b is of type float
  - c is of type complex
  - d is of type bool.

* Output :
  - [a+b] result in 15.5 because an integer and a float result.
  - [type (c)] displays the data type of variable c, which is complex.
  - [d] print True, as it is a boolean value.

Q.5] Evaluate relational logical & bitwise operations in python. Justify with suitable examples where each type of operator is most appropriate.

Relational Operators (>, <, ==, !=) are used to compare values and return boolean result. They are most appropriate in decision-making and conditional statements.

Q.6] Design & implement a menu-dirriven python program that, Accepts user input, performs arithmetic, relational, logical, and assignment operations, Display results clearly.

```
a= int(input("Enter frist number: "))
b= float(input("Enter secound number:"))

print("1. Arithmetic Operation")
print(" Addition:", a+b)

print("2. Relational Operation")
print(" a>b:", a>b)

print("3. Logical Operation")
print(" Both positive:", (a>0) and (b>0))

print("4. Assignment Operation")
  a += b
print(" Updated value of a:", a)
```

Q.7] Write a program in python to display " Hello World".

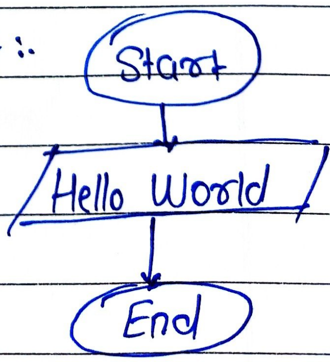
Algorithm:

1. Start

2. Display "Hello World"

3. Stop

Flowchart:



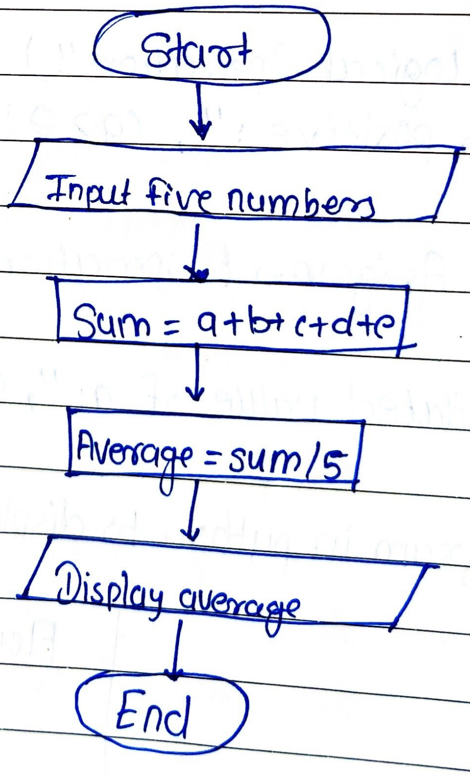
8. Write a program to perform average of five numbers using input() function and type conversion concept.

Print ("Hello World") $\xrightarrow{\text{Output}}$ Hello World"

8.

⇒ Algorithm :-
1. Start
2. Accept five numbers
3. Convert input to integers.
4. Calculate average
5. Display average
6. Stop

Flowchart :-



Program :-

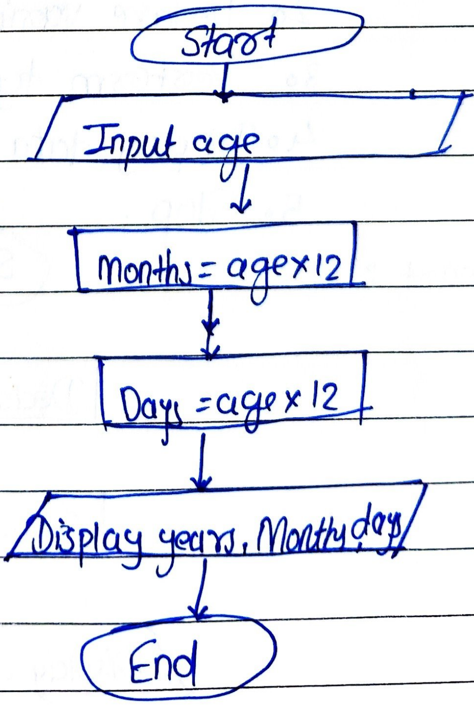
```
a=int(input("Enter number 1: "))
b=int(input("Enter number 2: "))
c=int(input("Enter number 3: "))
d=int(input("Enter number 4: "))
e=int(input("Enter number 5: "))

avg=(a+b+c+d+e)/5
print("Average: ", avg)
```

* Output :- Average : 20.0

Q. 9]

⇒ Algorithm :-
1. Start
2. Accept age in years
3. Convert years into months and days.
4. Display result
5. Stop

Flowchart :



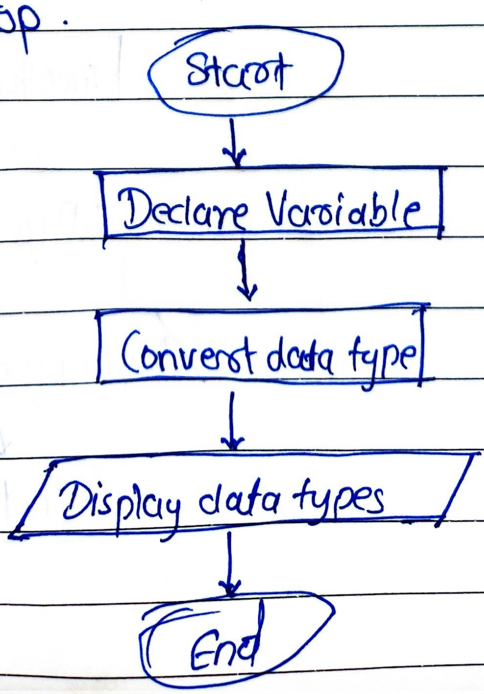
```
age = int(input("Enter your age in years:"))

months = age * 12
days = age * 365

print("Years:", age)
print("Months:", months)
print("Days:", days)
```

* Output :-> years : 20
Months : 240
Days : 7300.

Q.10]

=> Algorithm : 1. start
2. Declare variables
3. Perform type conversion
4. Display data types
5. Stop.

Flowchart :



Program :-

```
a= 10
b= 2.5
c = str(a)

print(type(a))
print(type(b))
print (type(c))
```

* Output :-

```
<class 'int'>
<class 'float'>
<class 'str'>
```